

Networks and Systems Management



Concept Sprint
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Management vs. Operations

Management:

High-level, conceptual aspects ensuring network performance

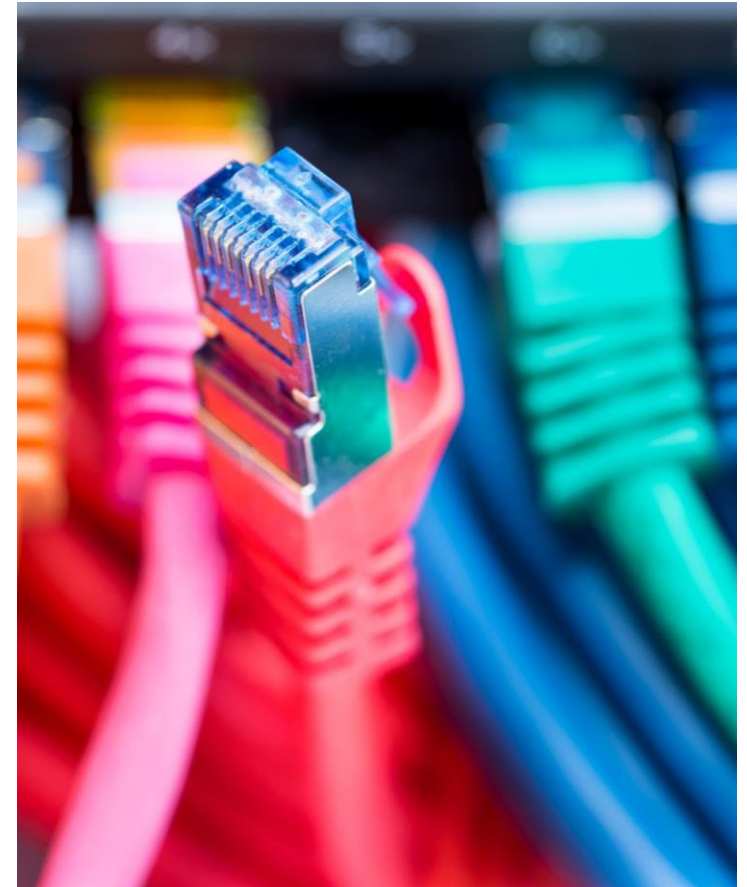
Key focus areas: **QoE, reliability, security**

Tasks: Monitoring, configurations, performance, fault detection, security, accounting

Operations:

Execution of management plans using people, processes, and tools

Typically handled in the **Network Operations Center (NOC)**



FCAPS Network Management Model

FCAPS Overview:

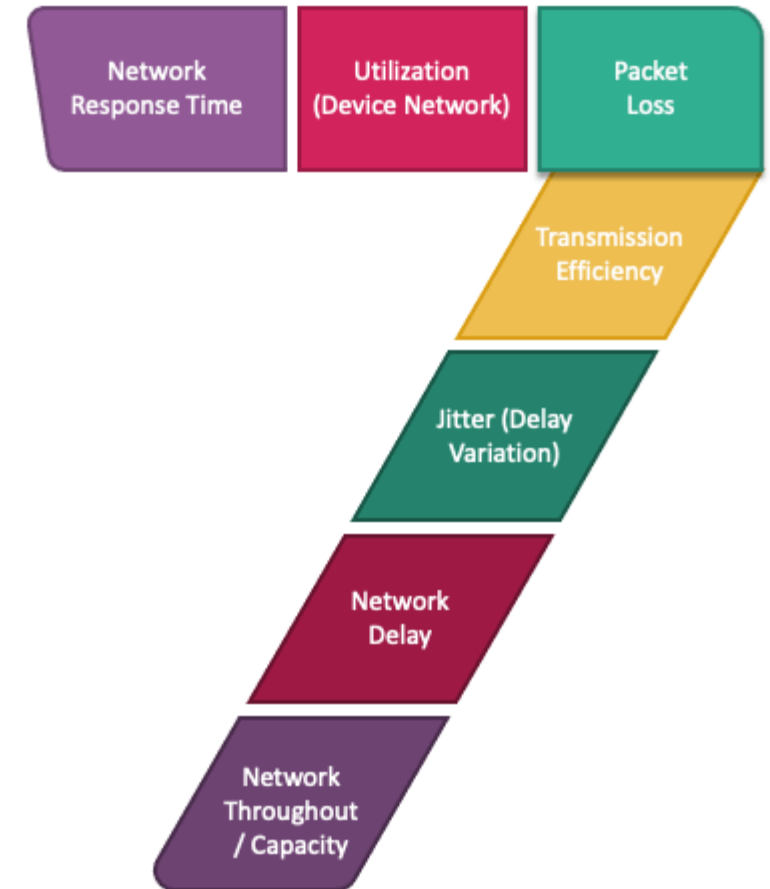
Fault Management: Detect, log, and resolve network failures

Configuration Management: Maintain and update device settings and network configurations

Accounting Management: Track resource usage for optimization and billing

Performance Management: Monitor network health and ensure optimal operation

Security Management: Protect network assets from threats and unauthorized access

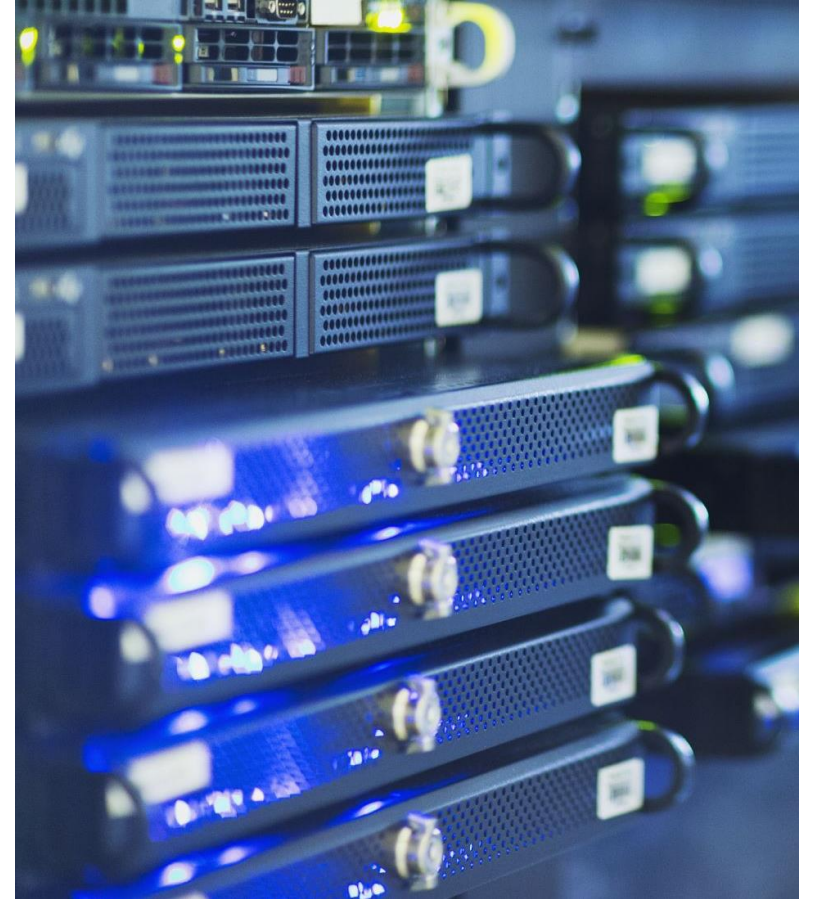


Dimensions of Networking - Packets

Data Plane: Microsecond-level packet forwarding & switching

Control Plane: Millisecond-level decisions (routing, QoS, firewall)

Network Services: DHCP, DNS, Radius, LDAP - mapping & authentication



Dimensions of Networking - Management

Configuration: Provisioning, updates, minor changes

Performance Monitoring: Ensure network meets SLAs

Fault Management: Detect & respond to failures

Security: Protect against attacks, unauthorized access



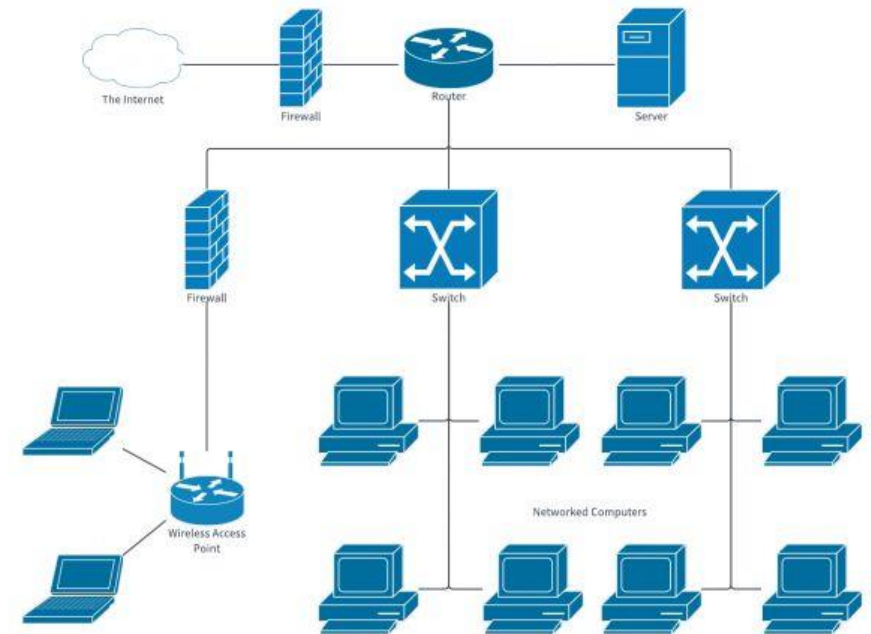
Changing Network Topologies

Spanning Tree Protocol: Prevents Ethernet loops

Fat Tree Topology: Optimized for data center bandwidth

Clos Network: Efficient multi-stage switching

North-South Traffic: User-to-Internet (Enterprise)
East-West Traffic: Server-to-server (Data Center)



Services & Systems

Traditional Services: Email, Web, Storage, Databases

Modern Systems: Microservices, REST APIs, Publish-Subscribe

Big Data Solutions: MongoDB, Hadoop, ELK

Virtualization & Containers: KVM, Docker, Kubernetes



DevOps in the Cloud

Traditional Deployment:

Long development cycles, strict separation between Dev & Ops

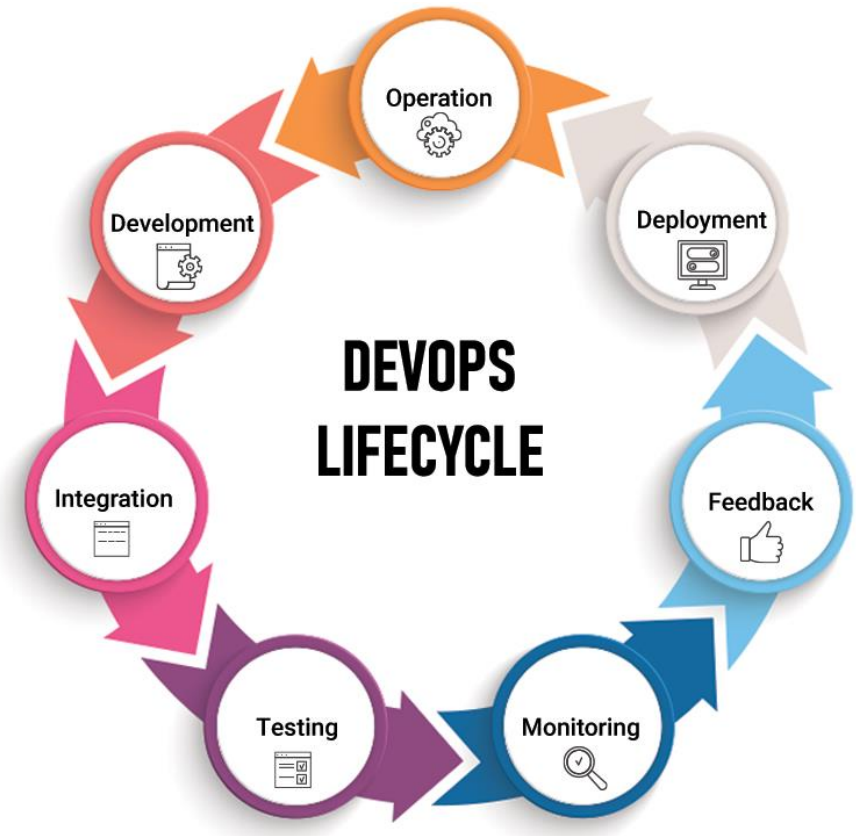
Delayed feedback, complex deployment processes

DevOps Approach:

Agile development, sprints, automation

Code built with deployment and testing in mind

Automated testing and deployment for reliability



DevOps for Networking

Challenges: Physical constraints, vendor APIs, complex protocols

Benefits:

- **Quick Deployment:** Automate provisioning & updates
- **Predictability:** Standardized configurations
- **Fail Fast:** Rapid rollback & recovery
- **Automation:** Reduce manual errors, increase efficiency



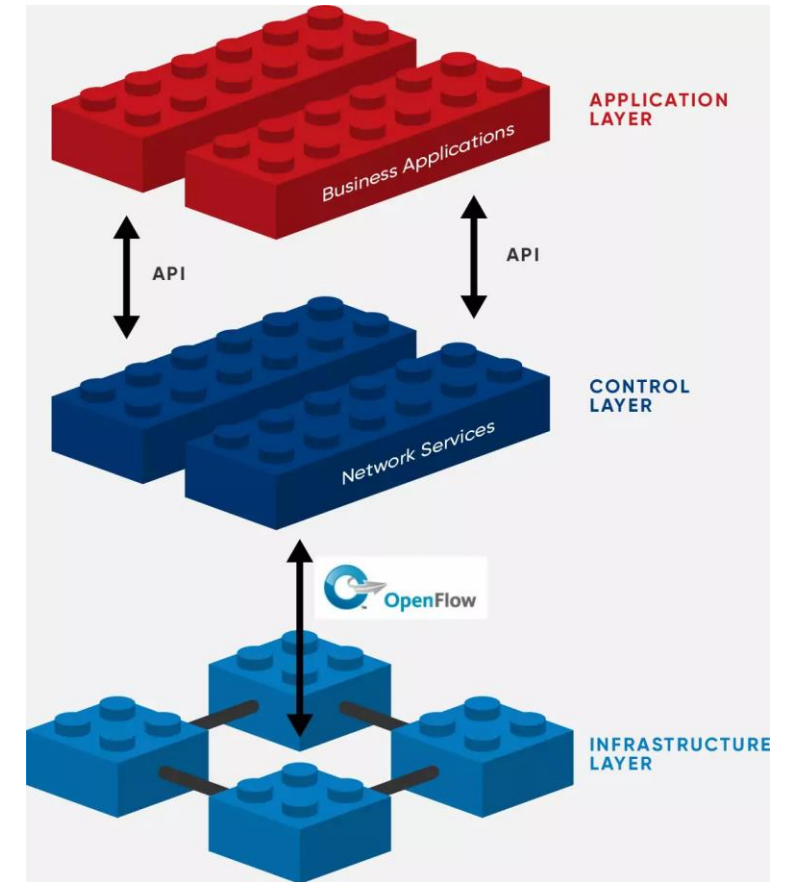
Moving Towards DevOps for Networking

Virtual Networks: VXLAN, connecting remote devices

Software-Defined Networking (SDN):
Centralized control over switch rules

Network Function Virtualization (NFV): Running network functions on generic hardware

Programmable Interfaces: API-driven network management



Key Takeaways

Management vs. Operations roles in networking

FCAPS framework for structured network management

Networking at different layers: packets, control, services

Evolution of network topologies for different traffic patterns

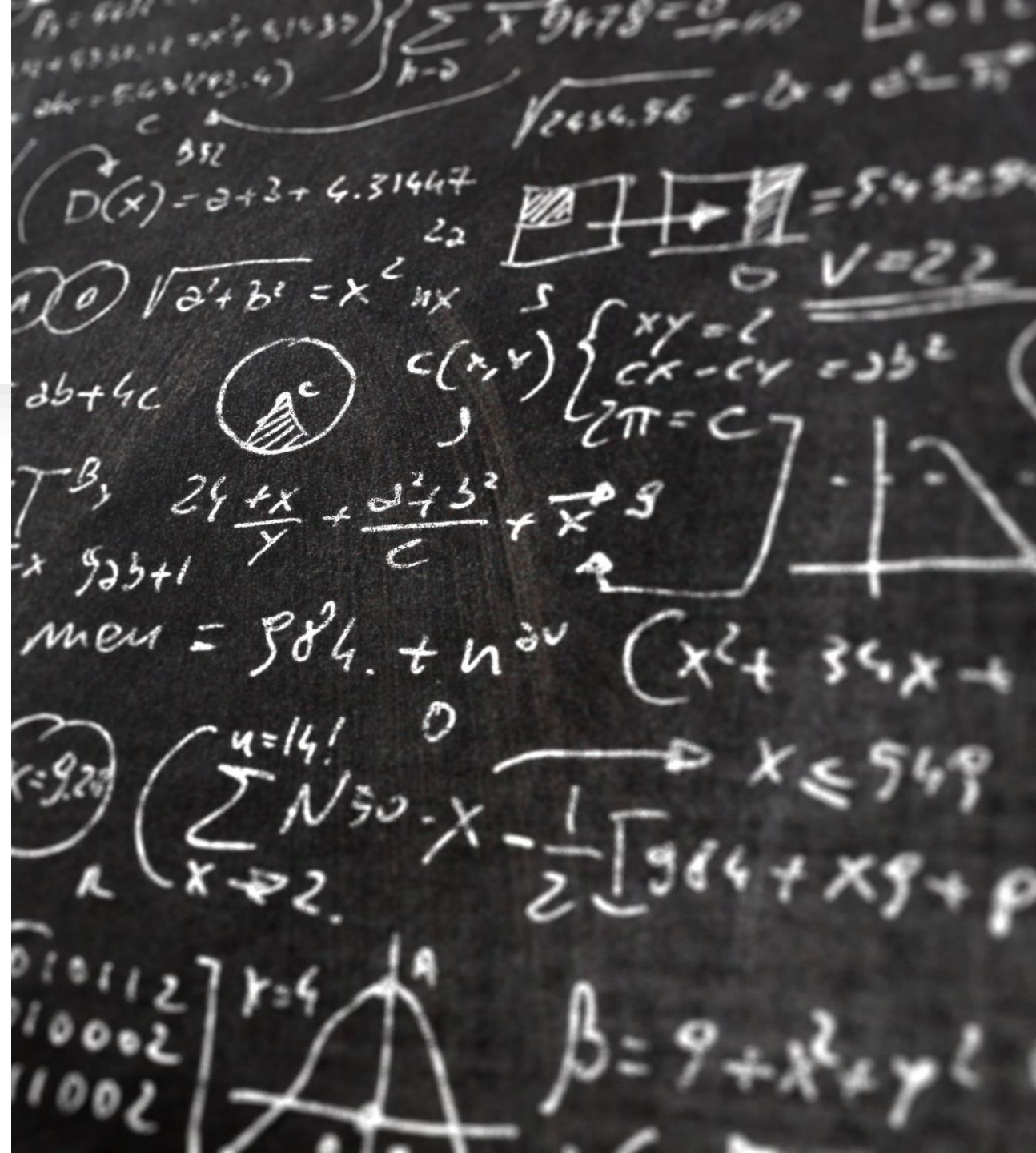
DevOps principles applied to cloud and networking

Automation & virtualization enabling modern network management



Further Reading & References

- [Cohen & Kaufeld \(1978\)](#) – NOC History
- [Das, Parulkar, McKeown \(2009\)](#) – Packet & Circuit Switching
- [Ayehu NOC Best Practices](#)
- [Kanwar & Raz \(2016\)](#) – Future of Network Operations



Discussion & Questions

What are key challenges in adopting DevOps for networking?

How can automation improve traditional network operations?

What are potential security risks in SDN and NFV?

How does FCAPS help ensure a reliable network infrastructure?

